

# MOHAMMAD RAMADAN ADAM

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## SUMMARY

AI Developer and R&D Engineer experienced in machine learning, computer vision, and LLM-powered applications. Skilled in building intelligent systems and translating AI research into scalable real-world solutions.

## PROFESSIONAL EXPERIENCE

### SEE, Cairo: Research and Development Engineer

Dec 2024 – Current

- Built an AI-powered chatbot integrated into the company website using a Retrieval-Augmented Generation (RAG) pipeline, enabling users to instantly access accurate information on product details, pricing, terms of use, and contact options — improving onboarding and reducing support requests.
- Developed advanced computer vision techniques to detect and analyze various physical training activities, including T-agility drills, running, push-ups, and other athletic exercises.
- Conducted in-depth research in the fields of depth analytics, pose estimation, and image enhancement, contributing to the development of robust AI models for athlete performance evaluation.
- Engineered a production-ready MLOps pipeline to deploy YOLO object detection models to AWS SageMaker, implementing cost-optimized auto-scaling to zero instances via CloudWatch-based scaling policies and inference components.
- Achieved a **94% infrastructure cost reduction** by architecting adaptive scaling strategies that automatically scale down idle resources, optimizing cloud spend without compromising model availability.
- Engineered AI-powered face recognition system for sports assessment videos using **AWS Rekognition** and **Python**, processing 6+ simultaneous athlete tracks with 90%+ accuracy threshold
- Designed priority-based identification algorithm with multi-tier tracking and confidence scoring, achieving **80% reduction** in cloud API costs through intelligent frame sampling and adaptive re-check intervals
- Developed a Text-to-SQL system that enables backend developers to query internal databases using natural language, improving data accessibility and reducing manual SQL query writing.

### Joe13, Remote: AI Developer

Apr 2024 – Nov 2024

- Developed an image similarity search model using image embeddings with CLIP and FAISS to retrieve similar images from a vector database, enhancing image search accuracy.
- Conducted research and experimentation on emerging trends and techniques in computer vision to improve model performance.
- Fine-tuned existing searching model and increased the accuracy by 7% leading to an increase in application performance.

### Click ITS, Cairo, EG: Data Scientist

Mar 2023 – Apr 2024

- Collaborated with government agencies to design and deliver AI-based solutions for specialized applications.
- Developed machine learning (ML) pipelines encompassing data extraction, cleaning, and model building using Dataiku and Huawei Mindspore.
- Established object detection and recognition models to enable precise readings from electric meters.
- Led a team of two data scientists in a project for the Ministry of Health to analyze hospital occupancies, using Dataiku's visualization and ML framework.
- Developed efficient data models to calculate hospital shortages and enhance operational efficiency.

## EDUCATION

### M.Sc., Data science

Queen's University  
Computer science

Graduated Mar 2023  
3.66 GPA

### B.Sc., Computer engineering

Arab Academy for Science, Technology and Maritime Transport  
Engineering

Graduated Aug 2021

## TECHNICAL SKILLS

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**Data Science:** Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, LLMS, RAG, Embeddings, Vector Database

**Languages:** Python, C++, Java, JavaScript, SQL

**Tools and Frameworks:** Tableau, Dataiku, TensorFlow, PyTorch, AWS SageMaker, Docker, FAISS, YOLO, CLIP

## PROJECTS

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### Football Analysis

[Link](#)

**Technologies used:** OpenCV YOLOv5 Pandas Numpy Ultralytics

- Designed and implemented a football analysis system utilizing YOLOv5 for object detection and tracking, enabling accurate, real-time tracking of players and the ball in video footage.
- Applied advanced computer vision algorithms for camera movement estimation, view transformation, and player-ball assignment, enhancing analysis accuracy and reliability.
- Engineered real-time video frame processing with added features like speed and distance estimation, team assignment, and visual annotations to provide detailed visual feedback during matches.
- Optimized workflow with efficient data handling techniques such as track interpolation and frame-wise processing, ensuring scalability and robustness for football analytics applications.

### AirBnB Price Classification

[Link](#)

**Technologies used:** LSTM GRU CNN Tokenization

- Designed and implemented a multi-modal model to classify Airbnb listings into price categories using combined text and image data.
- Applied NLP and image preprocessing techniques to handle text and visual input, enabling more accurate price categorization.
- Integrated advanced deep learning architectures including LSTM, GRU, and CNN for effective multi-modal analysis.

### AI-Powered Inventory Chatbot

**Technologies used:** OpenAI API Azure OpenAI Python Pydantic SQLite REST API HTTP Server HTML/CSS/JavaScript Prompt Engineering

- Developed a conversational AI chatbot that translates natural language queries into SQL commands, enabling non-technical users to query inventory databases intuitively through a web-based chat interface.
- Implemented dual LLM provider support (OpenAI and Azure OpenAI) with context-aware conversation management, session handling for multi-turn dialogues, and performance monitoring (token usage and latency metrics).
- Built a full-stack solution using Python with custom HTTP server and Pydantic models for type-safe API contracts, featuring a responsive web UI for real-time interactions.